SUKRITIN GUPTA

Data Scientist

Houston,TX | (203) 391-8555 | gsukritin@gmail.com | LinkedIn

SUMMARY

Dynamic and experienced Data Scientist with over 6 years of expertise in applying state-of-the-art technologies to develop impactful solutions. Specialized in leveraging Deep Learning, Machine Learning, and Natural Language Processing (NLP) to drive innovation and efficiency. Proven skills in utilizing advanced tools such as BERT, Transformers, and LSTM, consistently delivering models that yield quantifiable improvements. Adept in Python, Julia, and SQL, with extensive experience in building data pipelines, optimizing workflows, and solving complex problems. Notable achievements include the reduction of electricity costs by 14% in the oil & gas sector through pump head curve modeling and integration with power contracts. Holds a Master's degree in Data Science and is recognized for applying novel machine learning methods to generate actionable insights and contribute significant value to data-centric organizations.

WORK EXPERINCE

US Tech Solutions, TX (Enterprise Products) **Mar 2024- Present**

Data Scientist

- Developed and deployed scalable data pipelines using Python and Azure DevOps to map and monitor pump and delivery stations across an extensive oil & gas network in the US, ensuring accurate and real-time data flow reducing man hours by upto 40%.
- Engineered pump head curve models and integrated these with existing power contracts, resulting in a 14% reduction in electricity costs for pipeline operations.
- Created and optimized production-grade algorithms along with 200+ Pytest(s) to identify steady-state conditions and calculate DRA (Drag Reducing Agent) performance factors, enhancing pipeline efficiency and reducing operational costs.
- Led Git-driven version control and collaboration to streamline the development-to-production pipeline code, enhancing the efficiency of code deployment (27%) and operational workflows.
- Collaborated with cross-functional teams to align data pipeline outputs with business needs, contributing to strategic decision-making in energy optimization and pipeline management.

Teknobit, NJ **Aug 2023- Feb 2024**

Data Scientist

- Implemented a BERT Transformer multi-class classification model tailored for healthcare, extracting critical information from medical records and improving coverage by 11% compared to traditional regex-based pattern matching.
- Scaled solution on 900 million medical descriptions employing accelerated computing with GPU on AWS Sagemaker with PyTorch. Deployed model using Docker on AWS Sagemaker Inference and scheduled nightly runs using Airflow.
- Developed the business case and led proof-of-concept to predict patient appointment no-show probability using Random Forest in Python (scikit-learn) resulting into heightened efficiency and resource productivity for healthcare facilities, with an estimated annual revenue increase of approximately 7% ($14.5M).
- Launched a Data Layer Automation tool using Python and SQL for data mining and analysis of millions of rows of Clickstream data (>100GB) to support reporting and dashboard requirements of the product, leading to a reduction in dashboard load times of 75% and saved 8+ hrs. in creating new data layers for future reporting and dashboard needs
- Developed an XGBoost classification model for a major healthcare client with 88% accuracy and 0.94 AUC to predict HIV patient's persistence leading to an increase in patient retention by 15% quarterly and improve annual revenue by $1.8M by identifying key drivers behind patient churn
- Streamlined healthcare efficiency by teaming with 2 more resources to move 200 million entries into Fast Healthcare Interoperability Resources (FHIR) Store with US Healthcare standards, and designing seamless data communication.

CIRCANA (formerly IRI and NPD GROUP), CT **Jun 2022 – Jul 2023**

Data Scientist

- Developed and deployed predictive classification models, including Logistic Regression, SVM, Decision Tree, and Random Forest, resulting in a 15% reduction in Customer Churn Rate which translated to a revenue increase by 12%.
- Led the identification and resolution of control data contamination issues in A/B testing by implementing custom data-prep and modeling methodologies, leading to a substantial increase in lift from -37% to 16%.
- Leveraged a diverse range of models, including logistic regression, HLM, poison regression, and gamma distribution GLM, to drive data-driven insights and support decision-making processes.
- Resolved over 100 challenges within the Julia modeling pipeline, resulting in a significant 40% reduction in resolution time.
- Pioneered the introduction of a novel customer categorization metric based on mobile platforms (iOS/Android), reducing reliance on cookies and enhancing data accuracy.
- Achieved substantial efficiency gains by automating campaign read status reporting with Python, SQL and Pandas, reducing manual work hours from approximately 8 hours per week to just 2 hours per week.

National Grid, CT **Jul 2021 – Dec 2021**

Data Scientist

- Engineered a robust deep learning Recurrent Neural Network (RNN) model using Python to forecast electricity load in the US, achieving a remarkable accuracy improvement from 75% to 90%,
- Optimized data processing efficiency by implementing a streamlined code solution that automated data cleansing, sorting, and visualization tasks for data frames and Excel spreadsheets, leading to a 40% reduction in data processing time and saving $50,000 in operational costs annually.
- Spearheaded the automation of heatmap generation and visualization of gas consumption patterns across all US operations, facilitating detailed trend analysis by zip codes. This initiative led to a substantial 20% enhancement in operational efficiency.

New Holland, India **Aug 2017 – Aug 2018**

Project Management Engineer

- Conducted extensive data analyses, delivering actionable insights for 100+ datasets and achieving a 15% efficiency improvement. Collaborated with 10 teams, strengthening leadership and presentation skills.
- Administered 8 projects, saving $50,000 and ensuring on-time delivery, avoiding a 2-month delay. Implemented project management tools, saving 30 minutes daily. Monitored changes for projects exceeding $500,000 budget.
- Spearheaded market intelligence initiatives and collaborated with 5 business teams to successfully launch and market 3 new products. Resulted in a 20% increase in projected sales and contributed to achieving core objectives for each product launch.

ACADMICS PROJECTS

Microsoft & IBM stock analysis using Twitter Sentiment Analysis

- Technologies used: (Machine Learning, SAS, Python, Google Cloud Analytics, Data Pipelining via Airflow, Big Query, Natural Language Processing, Time-series, statistics).
- Identified 50% percent of stock market fluctuations connected to the predicted sentiments from the Twitter sentiment analyses, leading to a 9.2% improvement in investment returns

Texas Energy Demand Forecasting

- Leveraged long-term dependencies in electric load time series within Texas using RNN to reduce error rates of peak demand forecasting by over 20% and reduce the error rate of 24-hour ahead load forecasting by over 10%.

Capstone Project (Atlas Air)

- Employed predictive modeling in Python to optimize crew schedules, resulting in decreased crew dissatisfaction rate by 12% and increased crew retention rate by 10%, while taking into account schedule adjustment factors and crew dissatisfaction factors.

SKILLS

NLP: BERT, Transformers, LSTM, Recurrent Neural Networks, Sequence labeling, Machine Translation, Text Summarization, Intent Recognition, regex, Semantic Parsing, Entity Extraction, LDA, Text Mining, Information Extraction, Word2Vec, Natural Language Understanding, Natural Language Generation (NLG), Dialogue System, Question-Answering
Computer Vision: Object Detection, Image Segmentation, Image Classification, YOLO, CNN, ResNet, Inception
Deep Learning Libraries: TensorFlow, Keras, PyTorch, OpenCV, SparkNLP, Spacy, Gensim, Hugging Face
Machine Learning: Logistic Regression, Linear Regression, Decision Trees, Random Forest, XGBoost, Clustering, KNN, PCA, K-means, SparkML, Scikit-learn, A/B Testing, Statistics, Machine Learning, Hypothesis Testing, Data Mining, Supervised Learning, Unsupervised Learning
Programming & Data Manipulation: Python, R, PySpark, Pandas, NumPy, SciPy, Matplotlib, Plotly, Tableau
Database: SQL, PostgreSQL, Redshift, Big Query
Deployment Frameworks: GCP, AWS, AWS Sagemaker, Kubernetes, Docker, GitHub, Vertex AI, Elastic Search, Inverted Index, GPU, Fast API, Docker, S3, EC2.

EDUCATION

University of Connecticut Stamford, CT **Dec 2021**

Master of Science in Business Analytics and Project Management (Data Science)

GGSIPU, India **Jul 2017**

Bachelor of Technology in Mechanical and Automation Engineering

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals